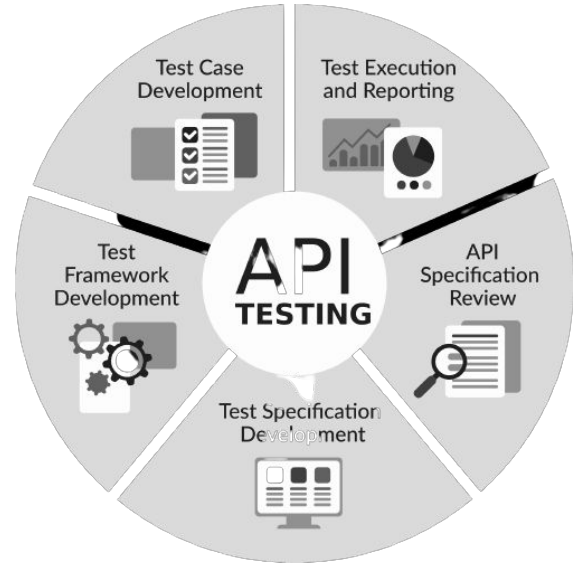


API Testing Mastery



Pramod Dutta
Lead SDET.



4x API Testing Projects

Agenda of Session

- Projects
- JS Overview, POSTMAN JS API
- Request Chaining
- POSTMAN Post Script Test
- Jenkins Overview
- Running POSTMAN collection in Jenkins
- Json Parsing
- JSON Schema Validation



Quiz

Which variable has the Highest priority?

- 1. Global**
- 2. Local**
- 3. Collection**
- 4. Environment**
- 5. Data (IterationData)**

401
Response



Quiz

What are the various authorization methods provided by Postman?

401
Response



Quiz

- API Key
- Bearer Token
- Basic auth
- Digest auth
- OAuth 1.0
- OAuth 2.0
- Hawk Authentication
- AWS Signature
- NTLM Authentication

401
Response



Quiz

Name some tools used for API Testing?

401
Response



Quiz

Postman
Katalon Studio
SoapUI
Tricentis Tosca
Apigee
Jmeter



Quiz

How do you set the same headers for all requests in a Postman Collection?

Right-click on the collection, navigate to the pre-request tab.

Add the below lines of code in the script to add a request header for all the requests present in the collection.

```
pm.request.headers.add({  
  key: 'TestHeader',  
  value: 'testValue'  
});
```

401
Response



Quiz

Consider that you get the following response to the request:

```
{  
  "timestamp": 1331856000000,  
  "status": 401  
}
```

401
Response

```
var jsonResponse = JSON.parse(responseBody); postman.setGlobalVariable("ts", jsonResponse.timestamp);  
var jsonResponse = JSON.parse(responseBody); postman.setEnvironmentVariable("ts", jsonResponse.timestamp);  
var jsonResponse = JSON.parse(responseBody); postman.setEnvironmentVariable("ts", timestamp.jsonResponse);  
var jsonResponse = JSON.parse(responseBody); postman.setEnvironmentVariable("ts", timestamp.jsonResponse);
```



Quiz

Which among the below options can be used to get the message from response headers?

`pm.getResponseHeader.message`

`pm.getResponseHeader(message);`

`pm.headers.message;`

`responseHeaders.message`

401
Response



Quiz

How do you access variable values from a file inside pre-request and test scripts?

`data.var_name`

`data['var_name']`

Both a and b

Variables cannot be accessed from file in the scripts.

401
Response



Quiz

Which is the correct form to insert an environment variable into a script in the tests tab?

`environment["var_name"]`

`{{var_name}}`

`EnvironmentVariable.var_name`

`pm.setEnvironmentVariable("var_name")`

401
Response



JavaScript Variables Revise

Request URL: <http://{{domain}}/users/{{userId}}>

Headers (key:value): X-{{myHeaderName}}:foo

Request body: {"id": "{{userId}}", "name": "John Doe"}



Assertions vs. Exception handling

- Assertions should be used to check something that should never happen,
- an exception should be used to check something that might happen.
- Example, a function might divide by 0, so an exception should be used, assertion could be used to check that the stack pointer is always ≥ 0
- An assertion would stop the program from running, but an exception would let the program continue running.
- **if an assertion fails it means that your code is wrong**



Javascript OverView

JavaScript Variables

JavaScript Data Types

JavaScript Functions

JavaScript Objects & Arrays

JS Conditional Statements If-Else, Switch, ForLoop

JS Cheatsheet - <https://sdet.live/9QWJ>



Javascript OverView

Practice

<https://jsbin.com/kekusuj/1/edit?js,console>

<https://jsbin.com/yujekuj/edit?js,console>



Parsing JSON

```
'{"name":"John", "age":30, "city":"New York"}'
```

```
const obj = JSON.parse('{"name":"John", "age":30,  
"city":"New York"}');
```

<https://www.getpostman.com/collections/780b145eb10cb133c397>



Quiz

Can global scope variables have duplicate names in postman?

True

False

401
Response



JavaScript libraries

```
responseHTML = cheerio(pm.response.text());  
console.log(responseHTML.find('[name="firstName"]'  
)val());
```



JavaScript Libraries

How to compare value of a response with an already defined variable?

```
// Getting values from response
var jsonData = pm.response.json();
var username = jsonData.userName;

// Saving the value for later use
pm.globals.set("username", username);
```

```
pm.test("Your test name", function () {
    var jsonData = pm.response.json();
    pm.expect(jsonData.value).to.eql(pm.globals.get("username"));
});
```

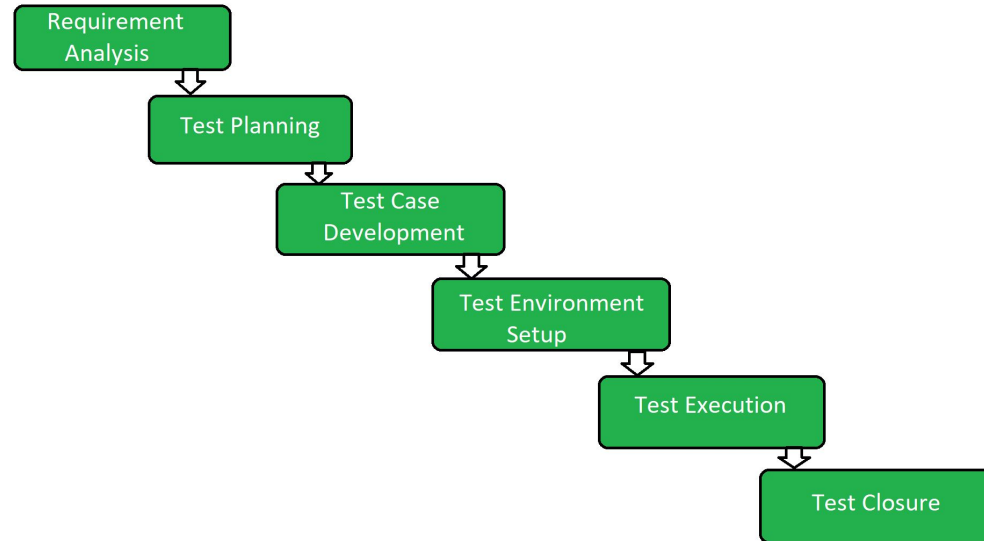


Test GITHUB Repo APIs



Manually Test APIs

STLC





Manually Test

Step1 :

Req Analysis - Understand the Req using API Doc

<https://docs.github.com/en/rest/repos/repos#create-a-n-organization-repository>

Confluence Doc -

<https://bugz.atlassian.net/l/cp/DgHWT9QL>



Manually Test

Step2 :

Test Plan - <https://sdet.live/testplan>



Manually Test

Step3 :

Test Cases - <https://sdet.live/tc-template>

<https://docs.google.com/spreadsheets/d/1m1fYVq5hAijYMJgP7iZESQDBkFihzqHw/edit?usp=sharing&oid=104755920778477387077&rtpof=true&sd=true>



Manually Test

Step4 : Test env

1. POSTMAN
2. Drive
3. Excel
4. CMD



Manually Test

Step5 : Test Execution & AUTOMATION

On POSTMAN.



Manually Test

Step6 : Closure

Test Report

1. Newman Execution
2. Manual execution report

<https://sdet.live/9ckA>



3rd Project GITHUB Repo APIs

1. GitHub Repo API Documentation.
2. CRUD OPeration on the Repo
3. Test cases of the GitHub Repo.
4. Writing POSTMAN Scripts.
5. Execution of the Scripts using Newman



3rd Project GITHUB Repo APIs

1. <https://docs.github.com/en/rest/repos/repos#create-an-organization-repository>
2. CRUD



3rd Project GITHUB Repo APIs

1. Get a repository.
2. Update a repository.
3. Delete a repository.
4. Create an organization repository



3rd Project GITHUB Repo APIs

Integration Scenario for the Github Repo API

Create -> Update -> Read -> Delete



LIVE CRUD in Postman #1



NewMan Commands

- `npm install -g newman`
- `newman run -h`
- Run collection -
`newman run mycollection.json`
- Run Collection with Env -
`newman run https://www.postman.com/collections/cb208e7e64056f5294e5 -e dev_environment.json`
- `-e` - environment, `-g` global , `-d` for iteration, `--f` folder, `-n` 3 iteration count.
- `--delay-request [number]` , `--timeout-request`
- `--bail` - Stop if any test fails
- `-k` for insecure



NewMan Reports

- **Newman-reporter-htmlextra**
 - `newman run collection.json -r htmlextra`
 - <https://github.com/DannyDainton/newman-reporter-htmlextra>



JIRA with Basic Auth

- <https://www.getpostman.com/collections/460b3c9cd7b62669cd97>
-



POSTMAN LIVE Projects

- <https://www.getpostman.com/collections/759cc00b76c41cdc9435>



Quiz

What are some of the JS libraries available in Postman?

401
Response

Thanks, for attending Class

**I hope you liked it. Say Thanks
in Comment :)**

Fin.